Skin Disease Detection

Using a Custom Convolutional Neural Network

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DATA FOUNDATION

The HAM10000 Dataset

- **10,015 images:** Diverse dermatoscopic dataset.
- **7 classes:** Covers melanoma, basal cell carcinoma, and benign lesions.
- **Challenge:** Significant natural class imbalance.
- **Pipeline split:** 80% train, 15% validation, 5% test.

MODEL DESIGN

Custom CNN Architecture

Built From Scratch

Zero pre-trained weights utilized.

Sequential Blocks

Stacked Conv2D, BatchNorm, and ReLU layers.

Robust Regularization

Max pooling and 0.5 dropout rate applied.

METHODOLOGY

Preprocessing & Training Strategy

Augmentation

Random flips, rotations, and color jitter applied.

Normalization

Images standardized using dataset mean and variance.

Optimization

Adam optimizer paired with active weight decay.

Early Stopping

Halted training dynamically to prevent overfitting.

EVALUATION

Final Model Performance

77.78%

Final Test Accuracy

0.932

Macro ROC-AUC

8.7M

Parameters Trained

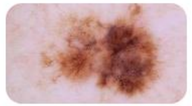
Conclusion

Successfully deployed a robust, scratch-built CNN pipeline capable of distinguishing complex skin lesions with high discriminatory power.

Thank You

Ready for Q&A

Image Sources



https://cdn.datasetninja.com/previews/q/ext:jpeg/resize:fill:400:0:0/q:70/plain/supervisely-assets-public/images/original/F/6/RM/CD74ZwL4UWjQYtTs4DmV1DpuMJoADjsBtWAlcVq5bnvlfq78kLuMugfIgqlulcnkzQsuUWIoU1gLC8NnFLBRGkSvdRtgQGOKnNOyCo5XN8w1fyk02DzibcnuNxFF.jpg

Source: datasetninja.com